1. Inferential Statistics

Inferential statistics involves using sample data to make predictions or generalizations about a population. It helps in drawing conclusions beyond the available dataset.

2. Hypothesis Testing

Hypothesis testing is a statistical method used to decide whether there is enough evidence to reject a null hypothesis. It involves null hypothesis (H_0), alternative hypothesis (H_1), test statistic, p-value, and significance level.

3. Confidence Interval

A confidence interval is a range of values used to estimate a population parameter. The critical value determines the width of the interval based on the confidence level.

4. P-value

The p-value represents the probability of obtaining results at least as extreme as the observed data, assuming the null hypothesis is true.

5. Type I vs Type II Error

Type I error occurs when a true null hypothesis is rejected. Type II error occurs when a false null hypothesis is not rejected.

6. Tests

- Z-test: Used when population variance is known and sample size is large.
- T-test: Used when sample size is small and variance is unknown.
- Chi-square test: Used for categorical data to test independence.
- ANOVA: Used to compare means across multiple groups.

7. Covariance

Covariance measures how two variables change together.

8. Correlation

Correlation measures the strength and direction of the relationship between two variables.

◈ 📄 Hypotheses Formulation

◈ **1. Hypothesis: Smoking vs BMI (T-Test)**

- **H_0 (Null Hypothesis):** There is no significant difference in BMI between smokers and non-smokers.

- **H_1 (Alternative Hypothesis):** There is a significant difference in BMI between smokers and non-smokers.

◈ 2. Hypothesis: Smoking vs Diabetes (Chi-Square Test)

- **H_0 (Null Hypothesis):** Smoking and diabetes are independent; smoking has no effect on diabetes prevalence.
- **H_1 (Alternative Hypothesis):** Smoking and diabetes are dependent; smoking affects diabetes prevalence.

◈ 3. Hypothesis: Age Group vs Glucose Level (ANOVA Test)

- **H_0 (Null Hypothesis):** There is no significant difference in mean glucose levels across different age groups.
- **H_1 (Alternative Hypothesis):** At least one age group has a significantly different mean glucose level.

◈ 4. Hypothesis: Age vs BMI (Correlation Analysis)

- **H_0 (Null Hypothesis):** There is no correlation between age and BMI.
- **H_1 (Alternative Hypothesis):** There is a correlation between age and BMI.